

Ensemble Algorithm Report

Introduction

In this project, machine learning techniques were applied using WEKA software to build a predictive model capable of forecasting the purchase decision (Purchased) based on several attributes such as Country, Age, Salary, and Gender.

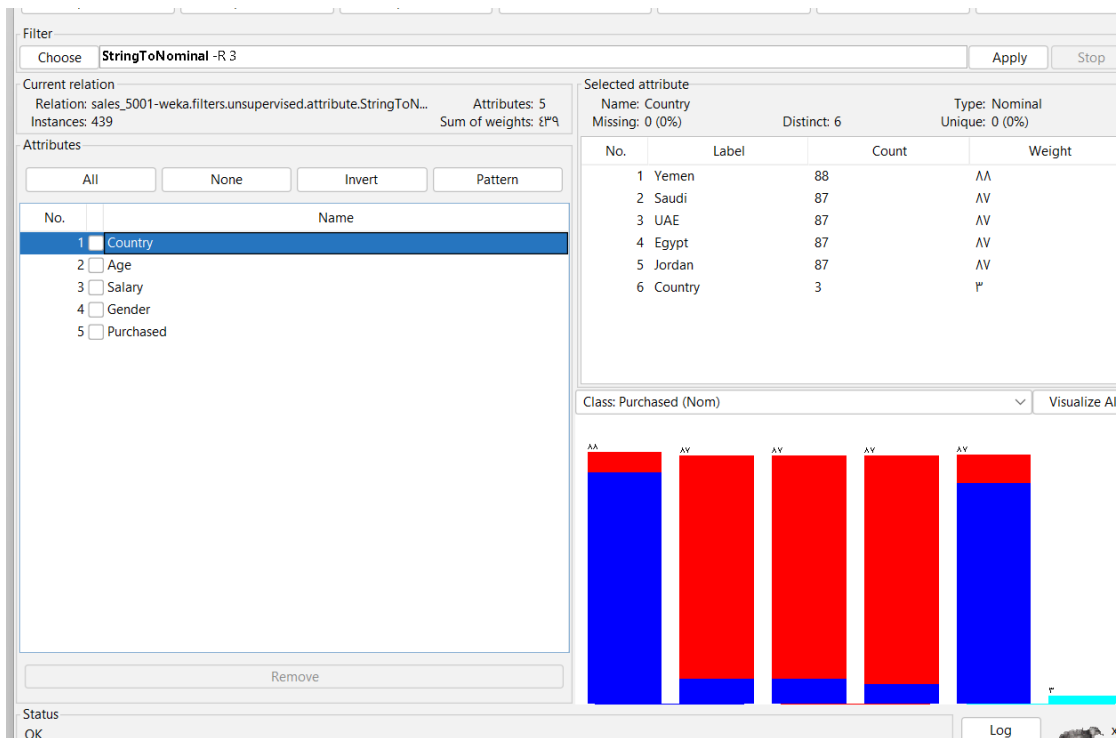
Multiple algorithms were used and then combined to achieve the best possible performance.

Dataset Description

Number of instances: 439 records

Number of attributes: 5

Problem type: Classification



Machine Learning Algorithms Used

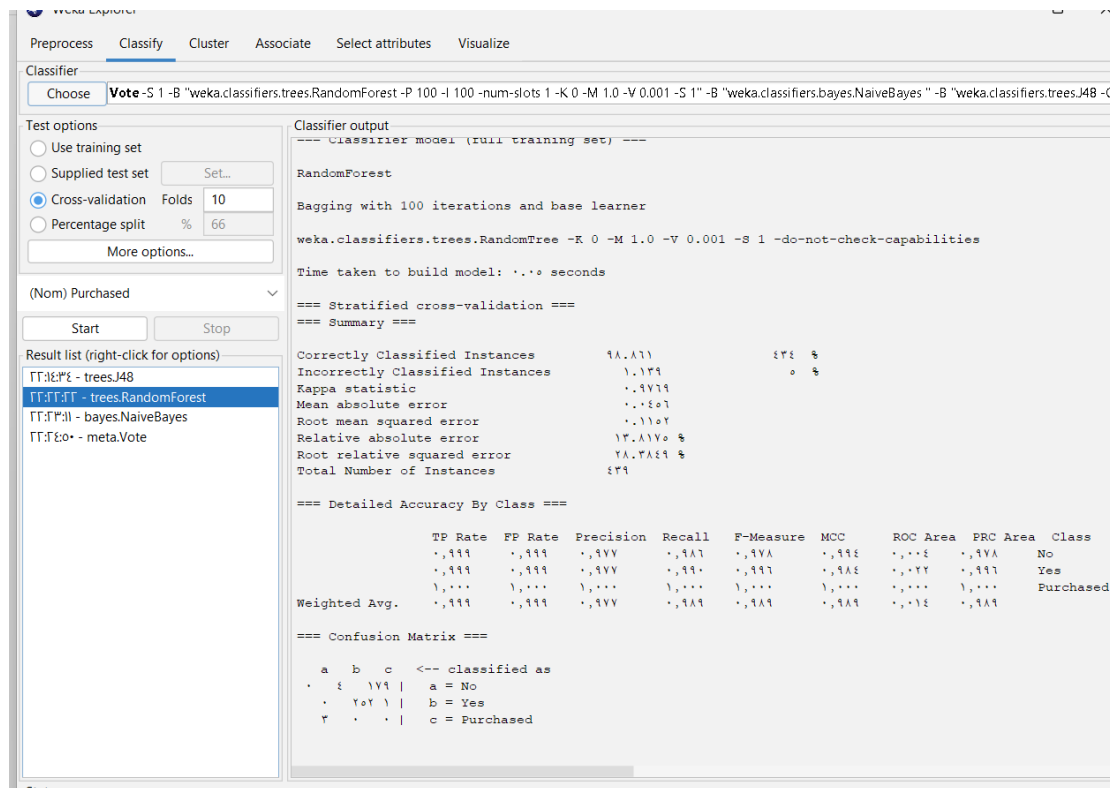
1. Random Forest

Random Forest is an algorithm that builds multiple decision trees instead of a single tree, and the final decision is made through majority voting.

- A large number of decision trees are created
- Each tree is trained on random subsets of the data
- Each tree produces a prediction
- The final output is selected based on majority vote

Advantages:

- High accuracy
- Reduces overfitting
- Stable and reliable results



2. Naive Bayes

Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem.

How it works:

- Calculates the probability of each class (Yes or No)
- Uses feature values to estimate probabilities

Selects the class with the highest probability

Advantages:

Very fast

Simple and efficient

Works well with small datasets

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'Vote - S 1 -B "weka.classifiers.trees.RandomForest -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1" -B "weka.classifiers.bayes.NaiveBayes" -B "weka.classifiers.trees.J48 -C'. The 'Test options' section shows 'Cross-validation' selected with 'Folds' set to 10. The 'Result list' on the left shows three classifiers: 'trees.J48', 'trees.RandomForest', and 'bayes.NaiveBayes'. The 'Classifier output' section displays the following data:

Classifier output			
gender			
Male	95.0	132.0	1.0
Female	90.0	123.0	1.0
Gender	1.0	1.0	4.0
[total]	186.0	256.0	6.0

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Metric	Value	%
Correctly Classified Instances	92.200	81.0 %
Incorrectly Classified Instances	21.800	19.0 %
Kappa statistic	0.812	
Mean absolute error	0.054	
Root mean squared error	0.114	
Relative absolute error	17.312 %	
Root relative squared error	12.912 %	
Total Number of Instances	241	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.949	0.000	0.812	0.902	0.880	0.931	0.849	0.880	No
	0.994	0.000	0.812	0.932	0.903	0.916	0.918	0.903	Yes
	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	Purchased
Weighted Avg.	0.991	0.000	0.812	0.922	0.922	0.922	0.888	0.922	

=== Confusion Matrix ===

	a	b	c	<-- classified as
a	22	11	0	a = No
b	21	12	0	b = Yes
c	0	0	6	c = Purchased

3. J48 (Decision Tree)

J48 is a decision tree algorithm used for classification tasks.

How it works:

Selects the best attribute for splitting the data

Divides data based on selected attributes

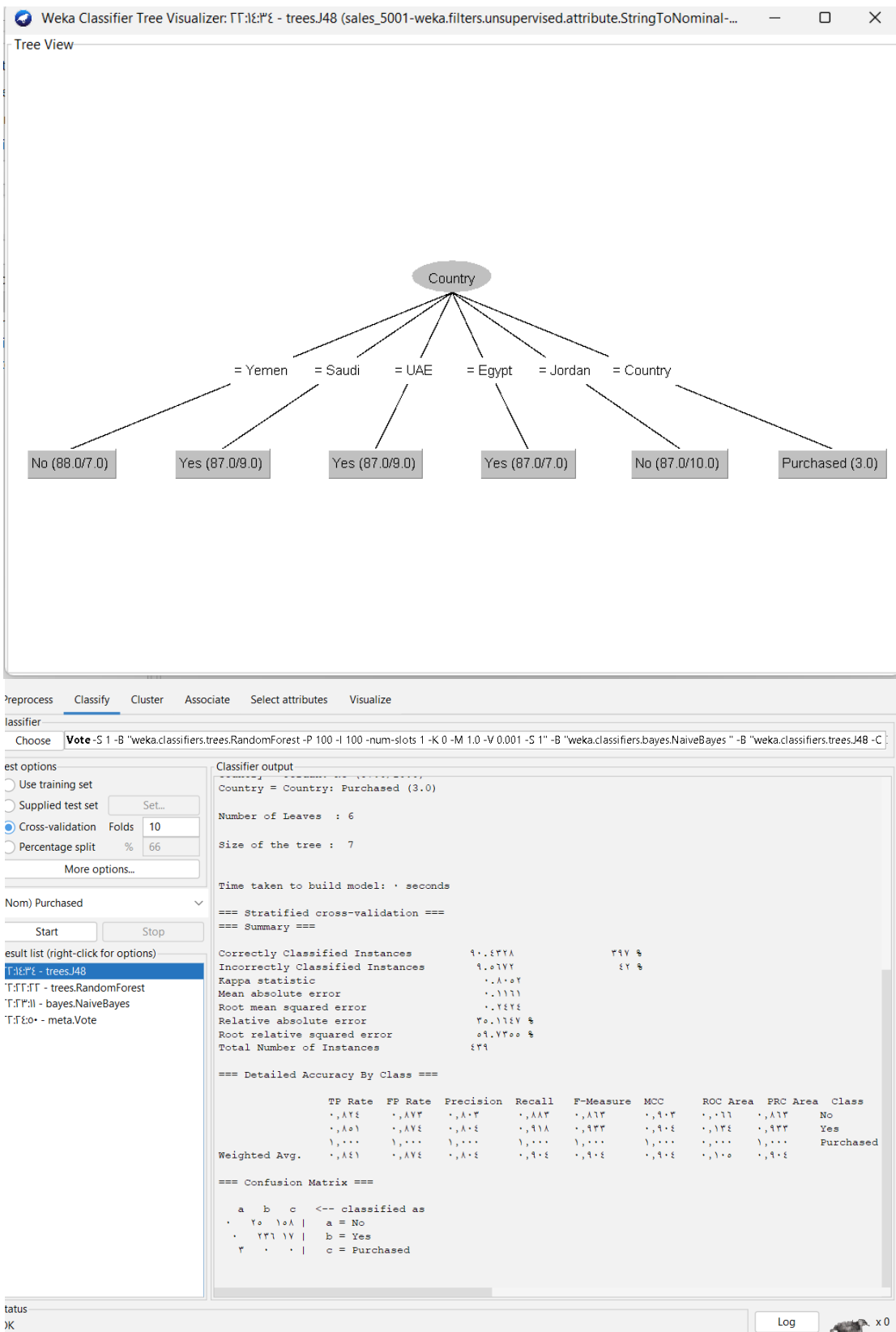
Repeats the process until reaching final decision nodes

Advantages:

Easy to understand

Produces clear decision rules

Good for interpretation and analysis



Ensemble Method (Voting)

The three algorithms were combined using a voting ensemble technique.

Concept:

Each algorithm produces a prediction

The results are averaged or voted on

The final decision is selected based on majority agreement

Objective:

Improve accuracy

Reduce prediction errors

Combine strengths of all algorithms

The screenshot shows the Weka GUI with the 'Classify' tab selected. The classifier chosen is 'Vote - S 1 -B "weka.classifiers.trees.RandomForest -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1" -B "weka.classifiers.bayes.NaiveBayes" -B "weka.classifiers.trees.J48 -C'. The test options are set to 'Cross-validation' with 'Folds' set to 10. The result list on the left shows 'TT:FT: - trees.J48', 'TT:FT:FT - trees.RandomForest', 'TT:FT:FT - bayes.NaiveBayes', and 'TT:FT:FT - meta.Vote'. The classifier output on the right shows the following summary:

```
==== Stratified cross-validation ====
==== Summary ====
Correctly Classified Instances      41.34%      401 / 489
Incorrectly Classified Instances    58.66%      88 / 489
Kappa statistic                    0.8238
Mean absolute error                 0.1762
Root mean squared error             0.4200
Relative absolute error             22.1219 %
Root relative squared error         51.7119 %
Total Number of Instances          489

==== Detailed Accuracy By Class ====
```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.9134	0.1100	0.8238	0.8238	0.8238	0.8238	0.9134	0.8238	No
	0.9134	0.1100	0.8238	0.8238	0.8238	0.8238	0.9134	0.8238	Yes
	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	Purchased
Weighted Avg.	0.9134	0.1100	0.8238	0.8238	0.8238	0.8238	0.9134	0.8238	

```
==== Confusion Matrix ====
  a  b  c  <-- classified as
+---+---+
+ 23 110 | a = No
+ 238 10 | b = Yes
+ 0 0 0 | c = Purchased
```

Model Results

Accuracy: 91.34%

Correctly classified instances: 401

Incorrectly classified instances: 38

Performance Analysis

Precision: ~91%

Recall: ~91%

F1-Score: ~91%

ROC Area: ~0.99

Decision Tree (J48) Analysis

The results showed that the **Country attribute** has the highest impact on the purchasing decision.

Findings:

Yemen and Jordan are mostly associated with “No purchase”

Saudi Arabia, UAE, and Egypt are mostly associated with “Purchase”

Reason for Selecting the Algorithms

Random Forest: High accuracy and reduced errors

Naive Bayes: Fast and probability-based approach

J48: Easy interpretation and rule extraction

These algorithms were combined to build a hybrid model that balances accuracy, speed, and interpretability.

Conclusion

A classification model was developed using three different machine learning algorithms and combined using a voting ensemble method. The final model achieved an accuracy of **91.34%**, demonstrating strong performance in predicting purchase behavior.